Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate cross correlation and autocorrelation on discrete time signals.**	
Experiment:-3	**Date:01-08-2026**	**Enrollment No:92400133057**

Github:-https://github.com/manavkasundra/DSIP

Objective:-
To understand the mathematical definitions of autocorrelation and cross-correlation.
To simulate these operations on deterministic and random discrete signals using Python.
To interpret the results for signal similarity, periodicity, and time-delay estimation.

Theory:-

What is Correlation?

Correlation is a mathematical way to measure how much two signals look alike as you slide one past the other to check for a match. It is widely used to find hidden patterns and calculate time delays in technologies like radar, sonar, and wireless communications.

What is Autocorrelation?
The autocorrelation function of a discrete-time signal $\boldsymbol{x[n]}$ measures the similarity between the signal and a time-shifted version of itself. For a finite-length signal of length $\boldsymbol{N}$, it is defined as:

$$\boldsymbol{r_{xx}[l] = \sum_{n=-\infty}^{\infty} x[n] \cdot x[n-l]}$$

where $\boldsymbol{l}$ is the lag (integer shift).
Interpretation:
At $\boldsymbol{l = 0}$, $\boldsymbol{r_{xx}[0] = \sum x[n]^2}$, which is the total energy of the signal.
Autocorrelation is even symmetric: $\boldsymbol{r_{xx}[l] = r_{xx}[-l]}$.
The maximum always occurs at zero lag: $\boldsymbol{| r_{xx}[l] | \leq r_{xx}[0]}$.

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Cross-Correlation:-

The cross-correlation between two different signals $\boldsymbol{x[n]}$ and $\boldsymbol{y[n]}$ measures how well $\boldsymbol{y[n]}$ matches $\boldsymbol{x[n]}$ when shifted:

$$\boldsymbol{r_{xy}[l] = \sum_{n=-\infty}^{\infty} x[n] \cdot y[n-l]}$$

Interpretation:-

If $\boldsymbol{y[n] = x[n-d]}$ (i.e., a delayed version of $\boldsymbol{x}$), then $\boldsymbol{r_{xy}[l]}$ will have its peak at $\boldsymbol{l = d}$.

Cross-correlation is not generally symmetric.

Normalized Correlation (Correlation Coefficient):-

Often we normalize to get values between -1 and 1 for easier interpretation:

$$\boldsymbol{\rho_{xy}[l] = \frac{r_{xy}[l]}{\sqrt{r_{xx}[0] \cdot r_{yy}[0]}}}$$

where $\boldsymbol{\rho = 1}$ means identical signals (for that lag), $\boldsymbol{\rho = -1}$ means opposite polarity, and $\boldsymbol{\rho = 0}$ means completely unrelated.

Applications in DSP:-

Time-delay estimation: Finding the lag where cross-correlation peaks gives the time shift between two received signals (e.g., in target ranging).

Periodicity detection: Autocorrelation of a periodic signal has peaks at multiples of the period.

Random signal analysis: White noise has an autocorrelation that is zero everywhere except at lag 0 (ideal for decorrelation).

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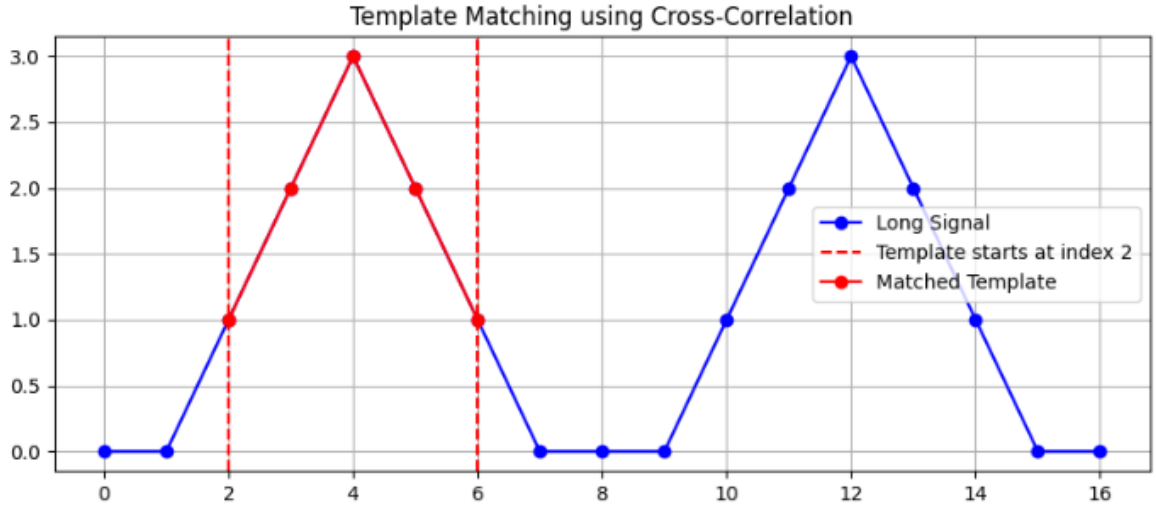
Normalized cross-correlation:-

Code:-

```
import numpy as np
import matplotlib.pyplot as plt
# Long signal
long_signal = np.array([0, 0, 1, 2, 3, 2, 1, 0, 0, 0, 1, 2, 3, 2, 1, 0, 0])
# Template to find
template = np.array([1, 2, 3, 2, 1])
# Compute cross-correlation
corr = np.correlate(long_signal, template, mode='valid')
peak_idx = np.argmax(corr)
plt.figure(figsize=(10, 4))
plt.plot(long_signal, 'bo-', label='Long Signal')
plt.axvline(peak_idx, color='red', linestyle='--', label=f'Template starts at index
{peak_idx}')
plt.axvline(peak_idx + len(template) - 1, color='red', linestyle='--')
plt.plot(np.arange(peak_idx, peak_idx + len(template)), template, 'ro-',
label='Matched Template')
plt.legend()
plt.title('Template Matching using Cross-Correlation')
plt.grid(True)
plt.show()
```

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Output:-



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Cross-Correlation to Detect Time Delay:-

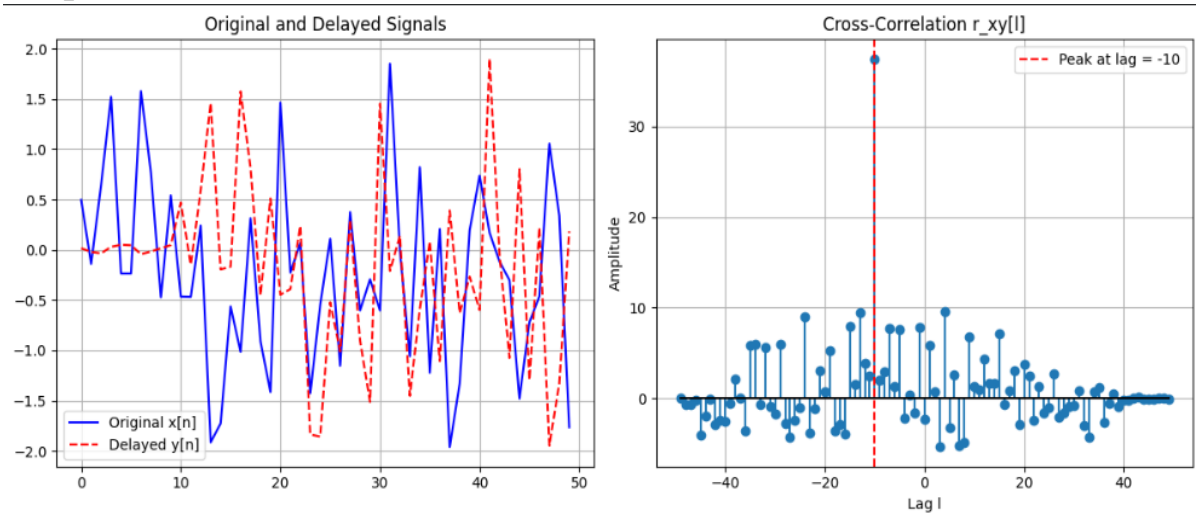
Code:-

```
import numpy as np
import matplotlib.pyplot as plt
# Generate a random signal
np.random.seed(42)
x = np.random.randn(50)
# Create a delayed version (delay = 10 samples)
delay = 10
y = np.zeros_like(x)
y[delay:] = x[:-delay]  # y[n] = x[n-10]
# Add a tiny bit of noise to make it realistic
y = y + 0.05 * np.random.randn(50)
# Cross-correlation
cross_corr = np.correlate(x, y, mode='full')
# Lags for cross-correlation: length = len(x) + len(y) - 1 = 99
Nx = len(x)
Ny = len(y)
lags = np.arange(-Ny + 1, Nx)
# Find the peak lag
peak_idx = np.argmax(cross_corr)
estimated_delay = lags[peak_idx]
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.plot(x, 'b-', label='Original x[n]')
plt.plot(y, 'r--', label='Delayed y[n]')
plt.legend()
plt.title('Original and Delayed Signals')
plt.grid(True)
plt.subplot(1, 2, 2)
plt.stem(lags, cross_corr, basefmt="k-")
```

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```
plt.axvline(estimated_delay, color='red', linestyle='--', label=f'Peak at lag =
{estimated_delay}')
plt.title('Cross-Correlation r_xy[l]')
plt.xlabel('Lag l')
plt.ylabel('Amplitude')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
print(f"True delay: {delay}")
print(f"Estimated delay from peak: {estimated_delay}")
```

Output:-




```
True delay: 10
Estimated delay from peak: -10
```

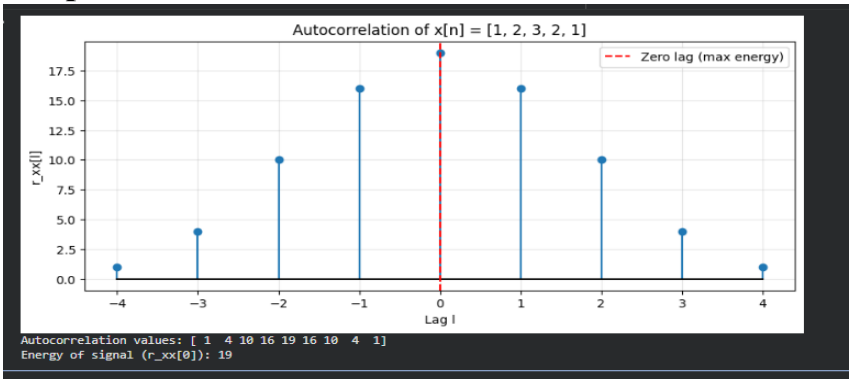
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Autocorrelation of a Deterministic Signal:-

Code:-

```
import numpy as np
import matplotlib.pyplot as plt
# Define a simple deterministic signal
x = np.array([1, 2, 3, 2, 1])
# Compute autocorrelation using numpy.correlate (mode='full' preserves all lags)
auto_corr = np.correlate(x, x, mode='full')
# Create lag axis. For length N, lags go from -(N-1) to (N-1)
N = len(x)
lags = np.arange(-N + 1, N)
# Plot
plt.figure(figsize=(10, 4))
plt.stem(lags, auto_corr, basefmt="k-")
plt.title('Autocorrelation of x[n] = [1, 2, 3, 2, 1]')
plt.xlabel('Lag l')
plt.ylabel('r_xx[l]')
plt.grid(True, alpha=0.3)
plt.axvline(0, color='red', linestyle='--', label='Zero lag (max energy)')
plt.legend()
plt.show()
print("Autocorrelation values:", auto_corr)
print("Energy of signal (r_xx[0]):", auto_corr[N-1])
```

Output:-



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Autocorrelation – Periodic vs. Random Signals:-

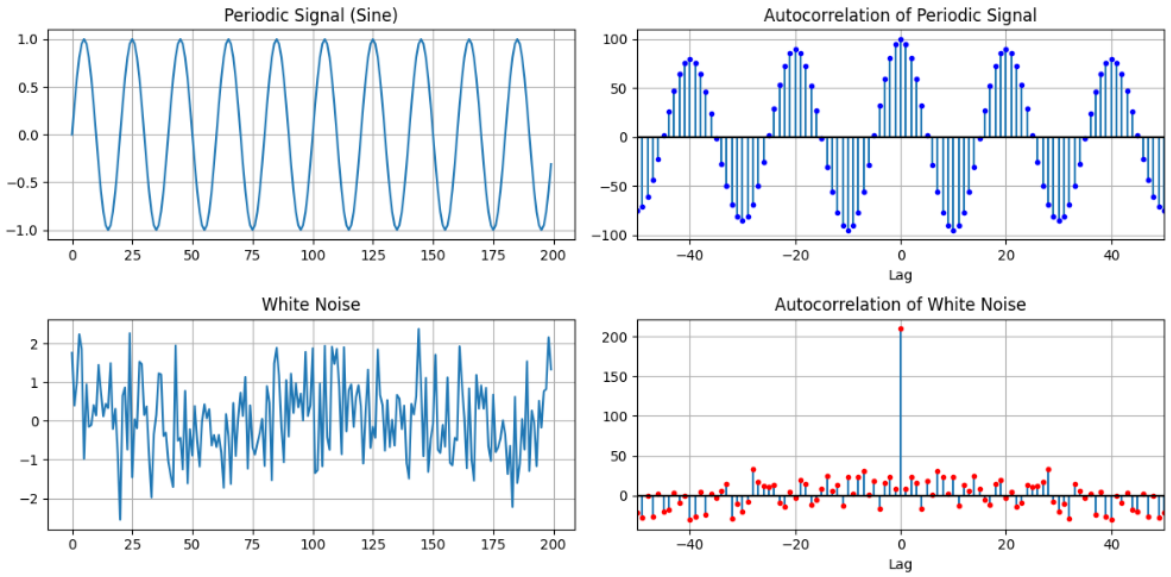
Code:-

```
import numpy as np
import matplotlib.pyplot as plt
n = np.arange(0, 200)
# Periodic signal: Sine wave
fs = 1
f = 0.05  # 5 cycles in 100 samples
periodic_signal = np.sin(2 * np.pi * f * n)
# Random signal (White noise)
np.random.seed(0)
noise_signal = np.random.randn(200)
# Compute autocorrelations
auto_periodic = np.correlate(periodic_signal, periodic_signal, mode='full')
auto_noise = np.correlate(noise_signal, noise_signal, mode='full')
N = len(periodic_signal)
lags = np.arange(-N + 1, N)
plt.figure(figsize=(12, 6))
plt.subplot(2, 2, 1)
plt.plot(n, periodic_signal)
plt.title('Periodic Signal (Sine)')
plt.grid(True)
plt.subplot(2, 2, 2)
plt.stem(lags, auto_periodic, basefmt="k-", markerfmt='b.')
plt.title('Autocorrelation of Periodic Signal')
plt.xlabel('Lag')
plt.xlim(-50, 50)
plt.grid(True)
plt.subplot(2, 2, 3)
plt.plot(n, noise_signal)
plt.title('White Noise')
plt.grid(True)
```

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```
plt.subplot(2, 2, 4)
plt.stem(lags, auto_noise, basefmt="k-", markerfmt='r.')
plt.title('Autocorrelation of White Noise')
plt.xlabel('Lag')
plt.xlim(-50, 50)
plt.grid(True)
plt.tight_layout()
plt.show()
```

Output:-



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Conclusion:-

Autocorrelation Characteristics: Comparing a signal to itself always produced a symmetrical graph with the highest peak at zero, which is a reliable way to measure its maximum energy.

Finding Time Delays: Cross-correlation proved to be a highly effective tool for calculating the time difference between two signals, finding the exact match point even when background noise was present.

Identifying Signal Types: The results made it easy to tell different signals apart. Repeating signals created wavy, repeating patterns on the graph, whereas random noise simply produced one sharp spike in the middle and stayed flat everywhere else.

Pattern Recognition: By using cross-correlation, we successfully located exactly where a small, known target pattern was hidden inside a much longer signal.